LAB 08

PROGRAMMING BASYS-3 BOARD

CONTENTS

WHAT IS AN FPGA?

- Field Programmable Gate Arrays (FPGAs) is a reconfigurable hardware chip consisting of a series of logic blocks which can be modified and configured by the user.
- FPGAs can be used to perform more specific and specialized tasks.
- Applications: Avionics, aerospace, military applications,
- These chips give the user much more flexibility and customization when performing specific tasks requiring **timely results**.
- FPGA's can be reprogrammed and implement custom hardware architecture.
- FPGAs are ideal for parallel systems where multiple tasks must be performed simultaneously. It is fast.

CPU VS FPGA

- **FPGA is application specific device and CPU is a general purpose device.**
- *FPGA:* Users can program the hardware configuration to meet particular requirements, making them ideal for specialized tasks such as signal processing, image processing, or real-time data analysis.
- *CPU:* CPUs are designed to handle a wide variety of tasks and are not optimized for any single application. They can run general-purpose operating systems and applications, making them versatile for everyday computing needs

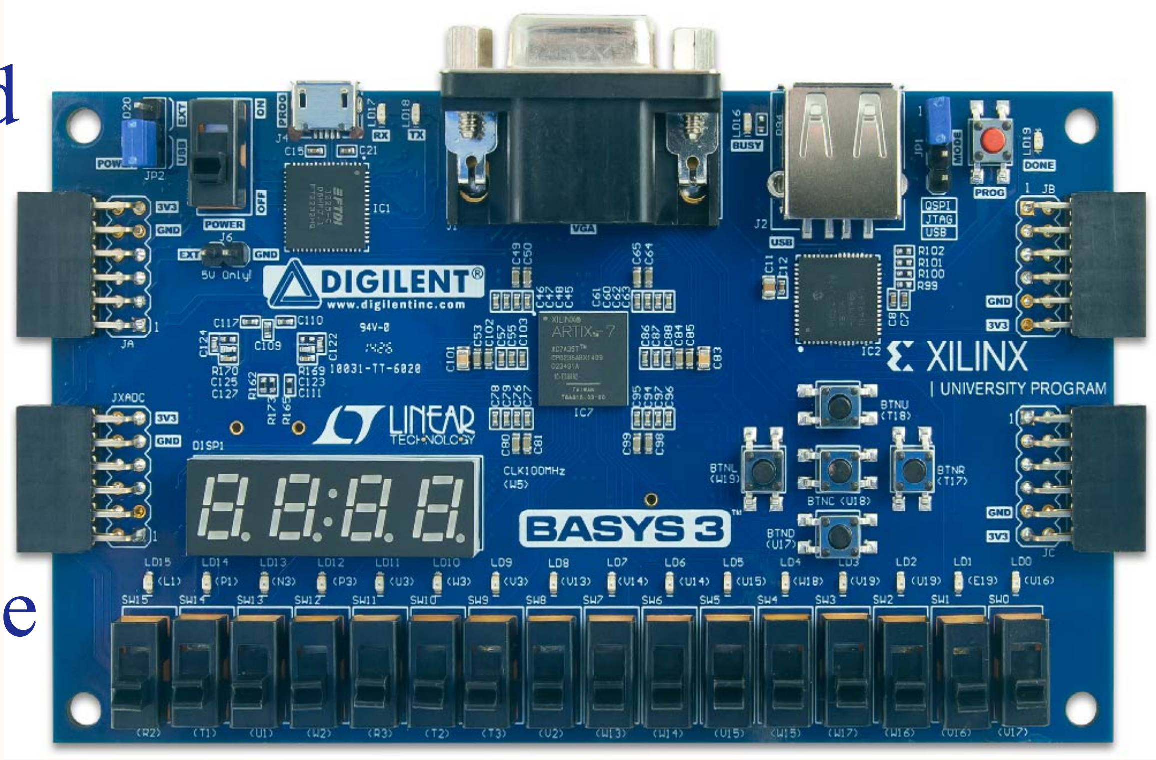
- **Speed:** FPGA is faster than CPU.

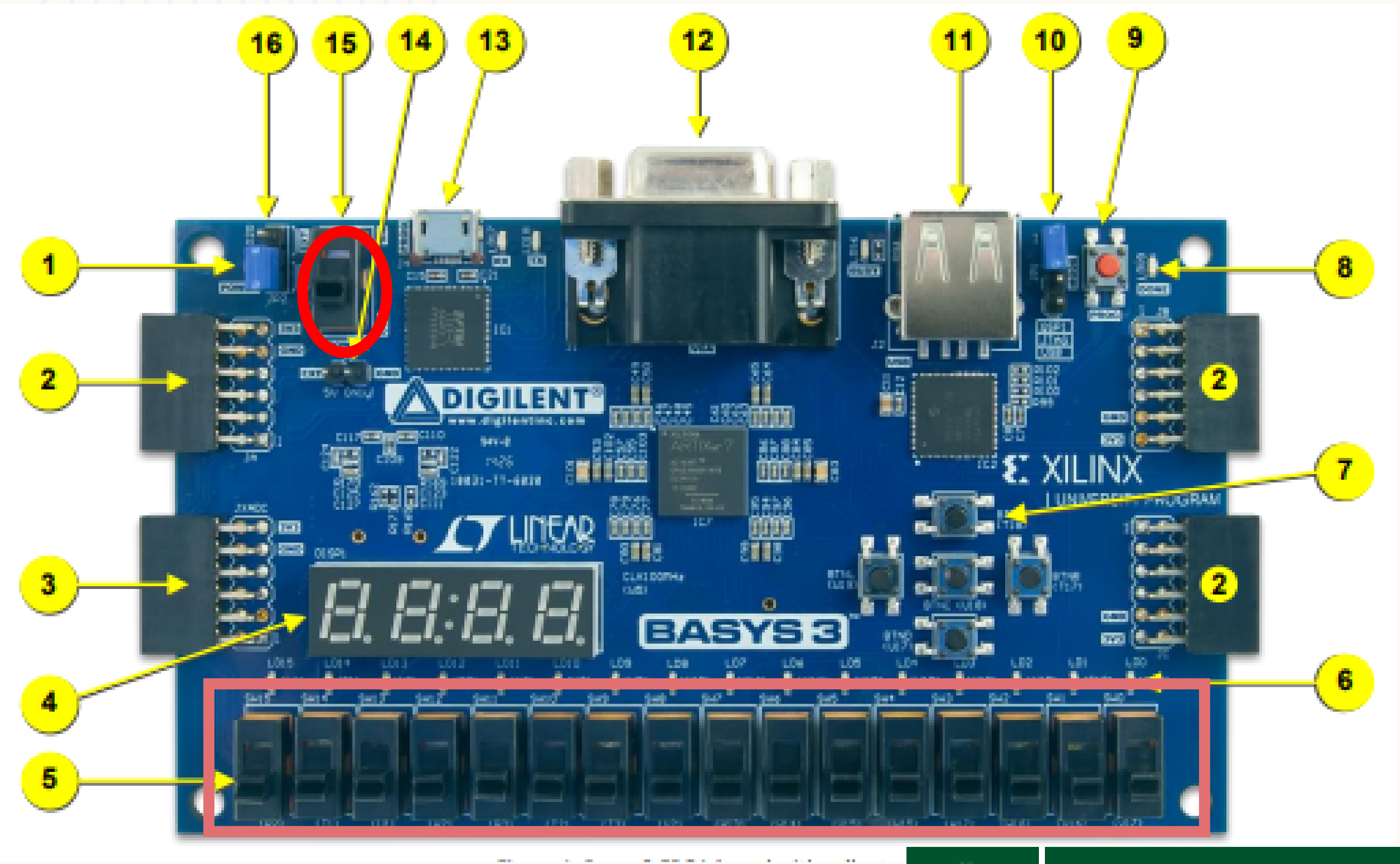
- **Power Consumption:** FPGA consumes more Power.

- **Price**: This is a tricky topic because both CPUs and FPGAs come in different sizes and speed, therefore the next statement should be taken carefully: CPUs tend to be cheaper than FPGAs.
- ***FPGA do not fit to mass production products due to their price.***

BASYS-3 BOARD

- Based on the latest Artix-7 processor.
- Includes a large collection of on-board I/O devices such as switches, LEDs, push buttons, seven segment display, and all required FPGA support components.
- Allows a large number of designs to be completed without the need for any additional hardware.





BASYS-3 BOARD

Callout	Component Description	Callout	Component Description
1	Power good LED	9	FPGA configuration reset button
2	Pmod port(s)	10	Programming mode jumper
3	Analog signal Pmod port (XADC)	11	USB host connector
4	Four digit 7-segment display	12	VGA connector
5	Slide switches (16)	13	Shared UART/ JTAG USB port
6	LEDs (16)	14	External power connector
7	Pushbuttons (5)	15	Power Switch
8	FPGA programming done LED	16	Power Select Jumper

BASYS-3 SOFTWARE COMPATIBILITY

The Basys-3 works with Xilinx's new high-performance Vivado™ Design Suite.

BITSTREAM

The FPGA configuration data is stored in files called bitstreams that have the .bit file extension.

TASK A & B

Task a:

- Create *AND module* (or use the module already created in Lab 05), & test your simulation.
- Attach screenshot of timing diagram of AND gate

Task b:

- Follow tutorial 3 (in the manual – also uploaded on LMS Canvas) to program Basys-3 board for *AND module* created in Task a.
- Attach snapshot of simulation output (all 4 outputs) of *AND gate* (Picture of your FPGA Board). The picture must not be blurry.
- *Demonstrate the programmed FPGA to RA.*

TASK C

- Use the *AddSub module* already created in Lab 07 & test your simulation.

- In design module of *AddSub module*, initialize ***4-bit output*** for seven-segment display activation. Send a low signal on one of them and keep rest high *to activate only one 7-segment display out of 4 of them*.

- For example: **assign seven-seg-display-active = 4’b1101** >> activates 3rd seven segment on the board from left [Active Low].

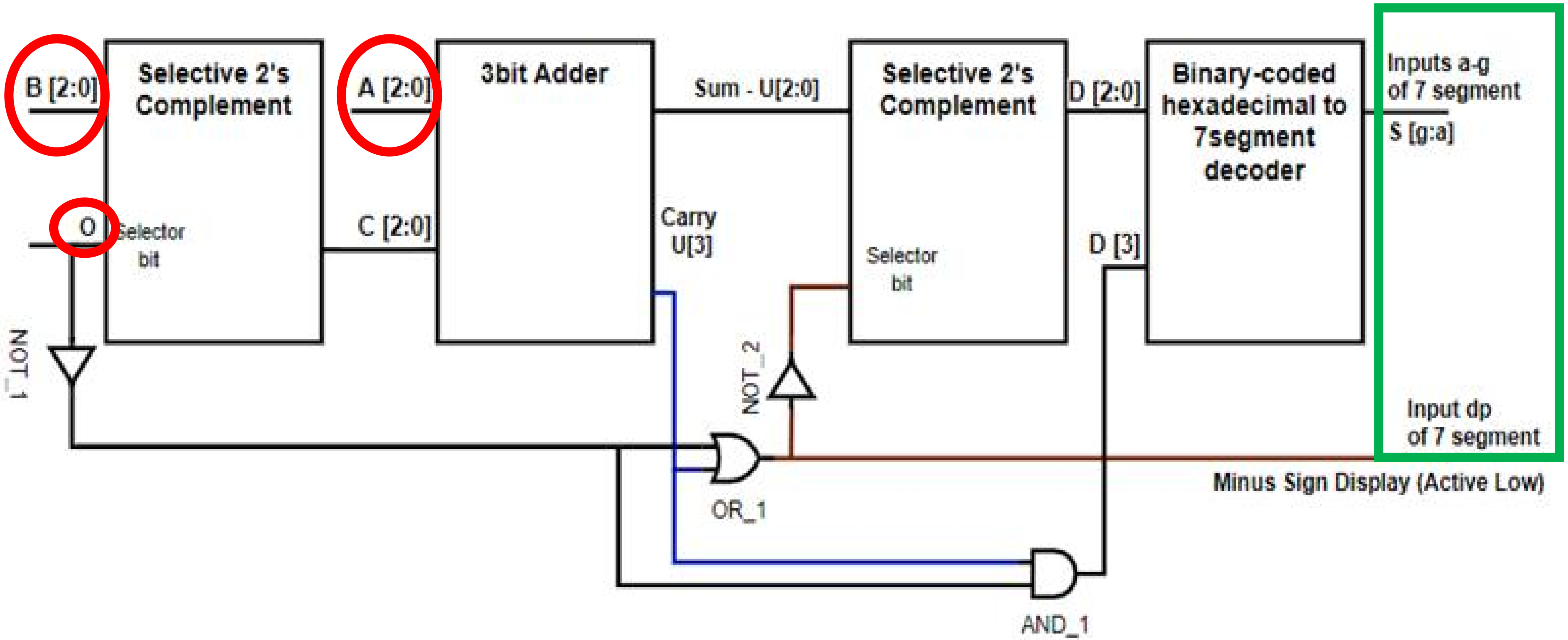
- Inputs *A(A2,A1,A0), B(B2,B1,B0)* and *O* are applied using slide switches – *7 switches are needed.*

- Output is displayed on one seven-segment display – *Use FPGA user manual uploaded on CANVAS for pin configuration.*

TASK C

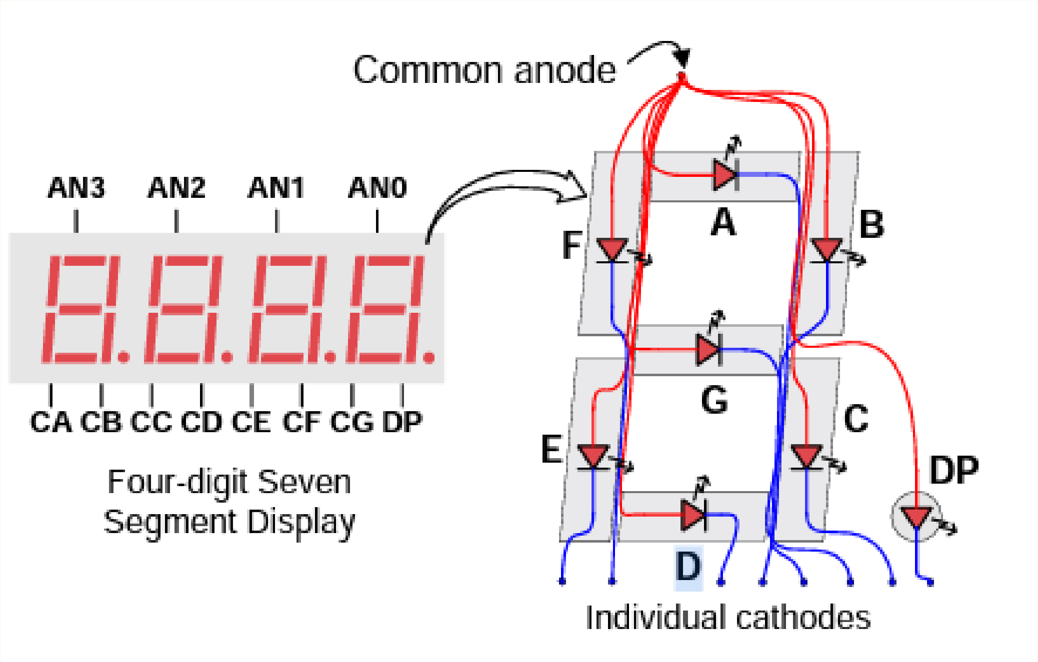
- Add codes of ***all*** design modules (testbench codes are not required).
- Add constraint file.
- Attach snapshot of simulation output (***at least 3 testcases – addition, subtraction with positive result, & subtract with negative result***) of *AddSub module* (Picture of your FPGA Board). The picture must be clear. Make sure to write the testcases and outputs details you choose to run.
- ***Demonstrate the programmed FPGA to RA.***

1 = Subtraction || Find 2's complement
0 = Addition || Pass through

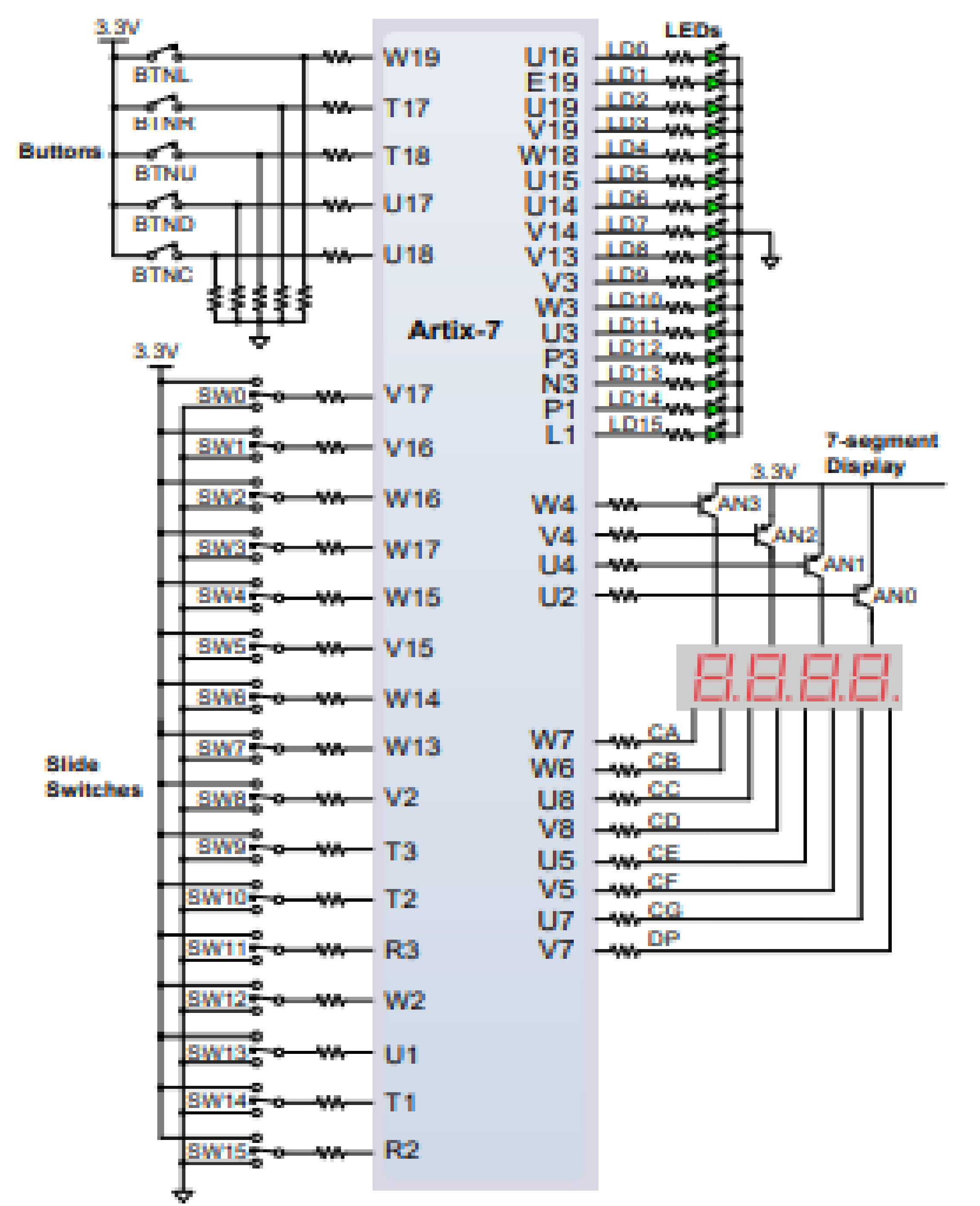


7-SEGMENT DISPLAY

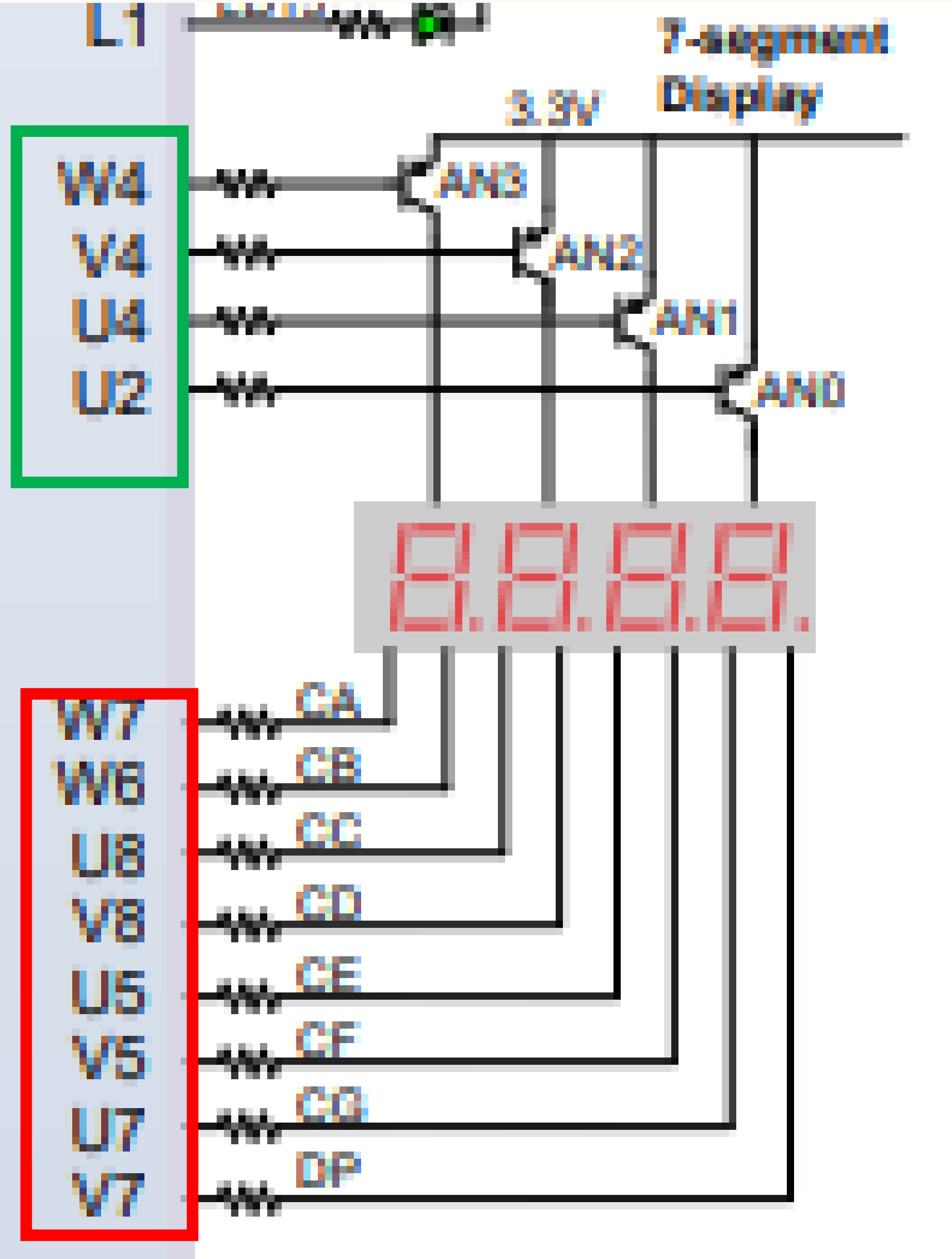
By-default the AN0, AN1, AN2 & AN3 pins getting low signal i.e., all 4 displays are in ON state.
Hence, adding another 4 bit output.



PIN CONFIGURATION OF FPGA



PIN CONFIGURATION OF SEVEN SEGMENT DISPLAY



- Input seven-seg-display-active *seg-active*:

seg-active[3] = W4
seg-active[2] = V4
seg-active[1] = U4
seg-active[0] = U2

- Output seven-seg *S*:
  CG = S[6] = U7
  CF = S[5] = V5
  CE = S[4] = U5
  CD = S[3] = V8
  CC = S[2] = U8
  CB = S[1] = W6
  CA = S[0] = W7

- Output seven-seg *DP*:

DP = dp = V7

- Watch video & Read user manual
- Return FPGA boards *with the right boxes* & cables once done